Formalização — Bruno Lopes (1240441)

Título

Spec-Driven for Agentic Software Development: A Characterisation and Comparison of Existing Approaches

Problema

AI coding agents now carry out complete development tasks: they analyse repositories, plan changes, modify multiple files and run tests. Spec-Driven Development (SDD) proposes that the specification should be the central artefact of this work, giving the agent persistent and structured context.

In little more than a year, several approaches have emerged that implement SDD in different ways, such as GitHub Spec Kit, OpenSpec, Kiro and the BMAD Method. They differ in the artefacts they require, the phases of the process and the points of human review.

A team that wants to adopt SDD has no comparable evidence today on which to base its choice. The available comparisons are almost all grey literature, and the academic works are conceptual, taxonomic or evaluate a single approach in isolation.

The problem to be addressed is the lack of a common characterisation of the existing SDD standards and of a comparison of their results when applied to the same project.

Quais as características do problema?

1. Envolve questões técnicas amplas ou de difícil harmonização?

Yes

The SDD approaches use processes, artefacts and tools that are not compatible with each other. Comparing them requires a common framework and an evaluation protocol that separates the effect of the approach from the effect of the agent, the model and the operator.

2. Tem uma solução óbvia?

No

Evidence on SDD is scarce and partly contradictory. Proponents report less rework, while a controlled study on repository context files found higher inference cost without an overall gain in task success. It is not known which approach pays off in which situation.

3. Aborda problemas não abrangidos pelas normas e códigos atuais?

Yes

There is no formal standard for specifications written for coding agents. The existing SDD approaches are de facto conventions defined by vendors or communities, and they change within months.

4. Envolve diversos grupos de partes interessadas (stakeholders)?

No

5. Inclui muitas partes componentes ou subproblemas, envolvendo múltiplas disciplinas?

Yes

The problem combines requirements engineering, software architecture, LLM-based coding agents, software quality and empirical research methods. It breaks down into identifying the standards, designing a reference project, instrumenting the experimental runs, analysing the results and building a decision guide.

6. Tem consequências significativas numa variedade de contextos?

No

Objetivos

Main objective: to identify and characterise the existing Spec-Driven Development standards for development with agents, and to compare their effect on the total cost and the quality of development when applied to the same project.

Specific objectives:

1. Identify the existing SDD standards and characterise them in a common comparative framework: what each one prescribes and how it can be used.
2. Design a reproducible evaluation protocol, with a reference project and cost and quality metrics defined in advance.
3. Apply each approach to the same example project, together with a baseline without a structured specification.
4. Compare the results of the approaches, with each other and with the baseline.
5. Produce a decision guide on which approach to use in which situation.

Research question: Which Spec-Driven Development approaches exist for development with agents, how do they differ, and how do they affect the total cost and the quality of development when applied to the same project?

Outcome atingível: Interpretação do problema a resolver

Haverá interpretação do problema a resolver?

Yes

The causes of the problem (the recent proliferation of SDD approaches without comparative evaluation), the stakeholders and their interests, and the consequences of choosing an approach without evidence are analysed. The problem is examined from several perspectives (cost, quality, human effort and platform dependence), confronting partly contradictory evidence from the literature.

Outcome atingível: Sistematização do conhecimento existente

Haverá sistematização do conhecimento existente para a resolução do problema?

Yes

A structured state of the art with a documented protocol: research questions, search strings, inclusion and exclusion criteria, and a PRISMA flow of the selection. The main sources are Scopus, Web of Science, IEEE Xplore and the ACM Digital Library. Tool documentation and repositories are included as primary sources for the approaches themselves, assessed with explicit quality criteria. The result is the identification of the existing SDD standards and a comparative framework describing what each one prescribes and how it can be used.

Outcome atingível: Avaliação de diferentes abordagens

Haverá avaliação de diferentes abordagens para a resolução do problema?

Yes

At two levels. In the state of the art, existing ways of evaluating SDD and coding agents (benchmarks, controlled experiments and industrial comparisons) are compared to justify the design of the evaluation. In the experiment, a set of SDD approaches selected with explicit criteria is applied to the same reference project and compared with each other and with a baseline without a structured specification.

Outcome atingível: Conceção de uma solução

Haverá conceção de uma solução (produto, componente, processo)?

Yes

A reproducible evaluation protocol is designed, including the reference project, a neutral requirements brief, acceptance tests, the experimental conditions, cost and quality metrics

defined in advance, and rules for human intervention. A decision guide is designed that relates SDD approach, type of task and context.

Outcome atingível: Construção de uma solução

Haverá construção de uma solução?

Yes

The reference project is implemented with each selected approach and with the baseline, using the coding agents each approach supports. The execution environment and scripts for automatic metric collection are built, and the artefacts are kept in a repository to allow replication. The decision guide is produced from the results.

Outcome atingível: Avaliação da solução

Haverá avaliação da solução desenhada/implementada?

Yes

The protocol is evaluated by its repeatability (variation between repeated runs of the same condition) and by its ability to distinguish the approaches. The decision guide is evaluated by applying it to tasks not used to build it and checking whether its recommendation matches the observed results. Threats to validity are discussed.

Referências bibliográficas

1. Böckeler, B. (2025). Understanding Spec-Driven-Development: Kiro, spec-kit, and Tessl. martinfowler.com. https://www.martinfowler.com/articles/exploring-gen-ai/sdd-3-tools.html
2. Díaz, J., Gayoso, J., Cimminio, A., Pérez, J. (2026). Spec-Driven Development for Agentic Software Engineering: Harnessing Human-Agent Teamwork. arXiv:2609.00252.
3. GitHub. Spec Kit. https://github.com/github/spec-kit
4. Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., Vechev, M. (2026). Evaluating AGENTS.md: Are Repository-Level Context Files Helpful for Coding Agents? arXiv:2602.11988.
5. Macedo, S. O. de (2026). From Prompt to Process: a Process Taxonomy and Comparative Assessment of Frameworks Supporting AI Software Development Agents. arXiv:2606.04967.
6. Panda, S. (2026). Citation Discipline in Spec-Driven Development: A Cross-Model Empirical Study of Output Determinism and Automated Hallucination Detection in LLM-Generated Code. arXiv:2606.30689.
7. Piskala, D. B. (2026). Spec-Driven Development: From Code to Contract in the Age of AI Coding Assistants. arXiv:2602.00180.